Alexandros Sigaras

Director, AI/XR,
Englander Institute for Precision Medicine
Assistant Professor of Research in
Systems and Computational Biomedicine

1305 York Avenue
Box 140
New York, NY 10021

+1 (646) 962-5634
als2076@med.cornell.edu
https://vivo.med.cornell.edu/display/cwid-als2076

Research Interests

My scientific focus is on building secure, scalable, and AI-driven computing solutions, largely in the cloud, that turn complex, multimodal medical data into actionable insight across precision oncology and beyond. This spans large-scale genomic analysis for cancer care alongside multimodal AI that integrates genomic, imaging, and bioacoustic data; through my role as Co-Investigator and Module Lead on the NIH Bridge2AI: Voice as a Biomarker of Health consortium, I work on ethically sourced voice datasets and foundation models that use the human voice as a biomarker of health. Complementing this, I lead research on extended reality (XR) and spatial computing in medicine, from immersive visualization of genomic and healthcare data for remote collaboration to AI-powered virtual humans and conversational agents for clinical training and education, uniting generative AI, large language models, and Mixed, Virtual, and Augmented Reality to reimagine how we explore data, learn, and deliver care.

Education

December 2013 — **M.S. in Computer Science**, *Columbia University*, Department of Computer Science, Fu Foundation School of Engineering and Applied Science, New York, NY, USA.
Track: Thesis Research Track
GPA: 3.851/4
Thesis: *Surgical Structured Light for 3D Minimally Invasive Surgical Imaging*

October 2011 — **B.S. in Informatics,** *University of Piraeus,* Department of Informatics, Piraeus, Greece.
Track: Information Systems Track
GPA: 8.5/10
Thesis: *Self Adaptive Robotic Warehouse Management Systems with event and location-based triggers*

Experience

Weill Cornell Medical College — **Assistant Professor of Research in Systems and Computational Biomedicine,**
Department of Systems and Computational Biomedicine

August 2025 - Present — **Director, AI/XR Lab**

Englander Institute for Precision Medicine

- Maintain, develop and manage efforts pertaining to the Laboratory for Artificial Intelligence and Extended Reality (AI/XR Lab) including leading the development of novel software applications in precision medicine.
- Lead the design and development of novel software solutions supporting the clinical and research activities at the Joint Clinical Genomics Initiative and at the Englander Institute for Precision Medicine.
- Oversee the design, development, and deployment of containerized solutions for the clinical genomics laboratory management system and data analysis, including QC/QA tools.
- Lead the efforts of large-scale genomic analysis in the cloud for both clinic and research activities; coordinate the development of artificial intelligence solutions to support the precision medicine knowledge base and the case review and sign out process.

Weill Cornell Medical College

April 2023 - Present

Director, Artificial Intelligence and Extended Reality (AI/XR)
Englander Institute for Precision Medicine

- Lead AI/XR research at the institution, establishing a research hub for AI/XR at the EIPM, as well as pursue funding opportunities for the institute's research initiatives.
- Develop and implement AI/XR strategies that align with the institution's overall goals and objectives.
- Manage the development and implementation of AI/XR projects, including managing budgets, timelines, and resources.
- Collaborate with clinicians, researchers, and students across Cornell campuses to identify opportunities for AI/XR to improve patient care, clinical workflows, and research outcomes, as well as develop a new syllabus to teach the future of medicine.
- Ensure the ethical use of AI applications, protecting patient privacy and avoiding biases that could lead to discrimination or harm.
- Stay ahead of the curve with AI/XR developments and research by attending conferences and staying abreast of emerging trends and technologies.
- Engage with external partners, such as industry partners and other academic institutions, to foster collaboration and identify opportunities for joint research and development projects.

Weill Cornell Medical College

April 2020 - Present

Assistant Professor of Research in Physiology and Biophysics,
Assistant Professor of Research in Computational Biomedicine,
Department of Physiology and Biophysics
Director, AI/XR Lab
Institute for Computational Biomedicine,
Englander Institute for Precision Medicine

- Maintain, develop and manage efforts pertaining to the Laboratory for Artificial Intelligence and Extended Reality (AI/XR Lab) including leading the development of novel software applications in precision medicine.

- Lead the design and development of novel software solutions supporting the clinical and research activities at the Joint Clinical Genomics Initiative and at the Englander Institute for Precision Medicine.
- Oversee the design, development, and deployment of containerized solutions for the clinical genomics laboratory management system and data analysis, including QC/QA tools.
- Lead the efforts of large-scale genomic analysis in the cloud for both clinic and research activities; coordinate the development of artificial intelligence solutions to support the precision medicine knowledge base and the case review and sign out process.

Weill Cornell Medical College

May 2017 – April 2020

Senior Research Associate in Computational Biomedicine, Department of Physiology and Biophysics Institute for Computational Biomedicine, Englander Institute for Precision Medicine

- Lead the design and development of novel software solutions supporting the clinical and research activities at the Joint Clinical Genomics Initiative and at the Englander Institute for Precision Medicine.
- Lead the development of the virtual and augmented reality program for precision medicine.
- Oversee the design, development, and deployment of containerized solutions for the clinical genomics laboratory management system and data analysis, including QC/QA tools
- Lead the efforts of large-scale genomic analysis in the cloud for both clinic and research activities; coordinate the development of artificial intelligence solutions to support the precision medicine knowledge base and the case review and sign out process.

Weill Cornell Medical College

April 2015 – May 2017

Research Associate in Computational Biomedicine, Department of Physiology and Biophysics Institute for Computational Biomedicine, Englander Institute for Precision Medicine

- Lead the search for new ways to leverage technology to advance precision medicine, including the use of Virtual Reality and Augmented Reality.
- Responsible for the design and implementation of the informatics infrastructure of the Englander Institute for Precision Medicine (EIPM), including the design and development of new tools to simplify user interaction with the laboratory information management system and the current IT infrastructure of WCM/NYP, collect and analyze user data and system performance within EIPM, and implement tools to report the results of the genomic analysis to the users via genomic portals.

Weill Cornell Medical College

March 2014 – April 2015

Programmer Analyst, Department of Pathology Institute for Computational Biomedicine, Englander Institute for Precision Medicine

	Software Development Lead for all clinical and R&D activities of the Englander institution for Precision Medicine. **Key Responsibilities:** Development of Next-Generation sequencing pipelines, LIS, cancer genomic portals and integrating genomic data to New York Presbyterian's HL7 infrastructure.
Columbia University	**Research Assistant (graduate), Robotics Lab**
September 2012 – March 2014	▪ **Project 1:** Developed prototype on Surgical Structured Light for 3D minimally invasive surgical imaging. ▪ **Project 2:** Developed brain computer interfaces for robotic grasping for people with locked-in syndrome.
Microsoft	**Internal Sales Education Specialist**
December 2011 – July 2012	Worked closely with universities and K-12 schools in Greece to provide cloud solutions and generally fulfill education institution needs in IT. Recipient of the Language Champion Award for the contribution to the Windows 8 International Review Program.
Microsoft	**Microsoft Student Partner**
November 2006 – December 2011	Member of Developers Platform Evangelists (DPE) group. Tasks included: administering the departmental Microsoft Developer Network Academic Alliance (MSDNAA) subscription, organizing technical presentations (for students) involving Microsoft products, advising students entering Microsoft's worldwide "Imagine Cup" programming contest, and setting up and moderating the studentguru.gr community website.

Support

Current Funding

NIH 1OT2OD032720-01 *"Bridge2AI: Voice as a Biomarker of Health - Building an ethically sourced, bioaccoustic database to understand disease like never before"* **Sigaras Alexandros (Co-I and Module Lead)**.
Duration: 09/2022 – 11/2026.

- Total costs for 2022 – 2023: $3,817,302.
- Total costs for 2023 – 2024: $5,302,032.
- Total costs for 2024 – 2025: $4,215,002.
- Total costs for 2025 – 2026: $4,660,942.

Honors and Awards

November 2015	**Nominee**, Forbes 30 under 30 in the Science and Healthcare categories

July 2012	**Language Champion Award** for Contribution to the Windows 8 International Review Program
June 2012	**Fulbright Scholar,** U.S Department of State's Bureau of Educational and Cultural Affairs
June 2012	**Scholarship** (for M.S. studies), Harry D. Triantafillu Scholarship Fund Award, Institute of International Education
October 2011	**Salutatorian**, University of Piraeus, Department of Informatics
October 2011	Graduated **summa cum laude**, University of Piraeus, Department of Informatics
November 2010	**Semi-Finalist**, Imagine Cup IT Challenge, Microsoft
December 2009	**Honorary Scholarship**, Hellenic American University
December 2009	**2nd place winner**, Athens Startup Weekend 2
November 2008	**Semi-Finalist**, Imagine Cup IT Challenge, Microsoft

Teaching

Teaching Assistant

- Created and graded homework, midterms, and final exams; prepared project material and gave lectures (Numbers in parentheses indicate enrollment).

COMS W4733 Computational Aspects of Robotics, Columbia University
Instructor: Prof. Peter Allen, Fall 2013 (60)

Certifications

December 2011	Udacity – Introduction to Artificial Intelligence - (top 10%)
October 2010	MCP, MCTS - 70-680: Microsoft Technology Specialist – Windows 7, Configuration
June 2007	MCP, MCTS - 70-680: Microsoft Technology Specialist – Windows Vista, Configuration

Professional Service and Academic Leadership

Scientific Review Boards and Committees

- Member, Scientific Review Committee (SRC), Weill Cornell Medicine-Qatar (WCM-Q) , Doha, Qatar May 2025 – Present
    - Review Editor for “Virtual Reality in Medicine” – Journal of Frontiers in Virtual Reality Guest Editor for “XR in Surgical and Procedural Medicine” – Journal of Medical Extended Reality

Editorial and Peer Review Activities

- Review Editor for “Virtual Reality in Medicine” – Journal of Frontiers in Virtual Reality Guest Editor for “XR in Surgical and Procedural Medicine” – Journal of Medical Extended Reality
- Guest Editor Assistant for “Clinical Medicine and Surgery in the Digital Age: Advances in Virtual, Augmented, Mixed, and Extended Reality” – Journal of Clinical Medicine
- Reviewer – Nature

Publications

Journal Articles

[J.40] Elemento O, **Sigaras A**, Colonel J, Hajirasouliha I, Ghosh S, Bensoussan Y, Bridge2AI-Voice Consortium, Rameau A. Towards A Foundation Model for Clinical Voice Biomarkers. ***medRxiv***, 2026.05.28.26354346; doi: https://doi.org/10.64898/2026.05.28.26354346

[J.39] Waltman P, Chandra P, Eng KW, Wilkes DC, Park H, Pabon C, Delpe P, Bhinder B, Manohar J, Kane T, Fernandez E, Gorski K, Greco N, Simi M, Tang JM, Zisimopoulos P, King A, Al Assaad M, Ten Eyck T, Roberts D, Monge J, Demichelis F, Tam W, Ouseph MM, **Sigaras A**, Beltran H, Rennert H, Lindeman N, Song W, Solomon J, Mosquera JM, Kim R, Catalano J, Hassane DC, Sigouros M, Elemento O, Alonso A, Sboner A. EXaCT-2: an augmented and customizable oncology-focused whole exome sequencing platform. ***NPJ Precis Oncol.*** 2026 Apr 21. doi: 10.1038/s41698-026-01390-5. Epub ahead of print. PMID: 42014463.

[J.38] Kozan EN, Delgado-de la Mora J, Al Assaad M, Ginter P, Gundem G, Levine MF, Gorski K, Phan A, Beg S, **Sigaras A**, LaPolla D, Semaan A, Wilkes DC, Greco N, Sigouros M, Manohar J, Medina-Martínez JS, Sboner A, Elemento O, Hoda S, Saxena A, Andreopoulou E, Boyraz B, Mosquera JM. Whole genome sequencing of locally advanced and metastatic breast carcinoma unravels relevant molecular signatures and novel events. **Pathol Res Pract.** 2026 Jun;282:156469. doi: 10.1016/j.prp.2026.156469. Epub 2026 Apr 15. PMID: 42013747.

[J.37] Henry O, Brumberger E, Jotwani R, **Sigaras A**, Sriram S, Yeh IM, Rubin J. Using an Artificial Intelligence Conversational Agent in Virtual Reality to Teach Serious Illness Communication Skills to Residents. ***J Grad Med Educ.*** 2026 Apr;18(2):193-194. doi: 10.4300/JGME-D-25-00952.1. Epub 2026 Apr 15. PMID: 42005895; PMCID: PMC13086130.

[J.36] He L, Li H, Wang S, Baik E, Kervin S, Zhao R, Ramos JM; **Bridge2AI-Voice Consortium**; Colonel J, Rameau A. Decoding Gender in Cough Sounds: A Transformer-Based Analysis. ***Laryngoscope***. 2026 Jan 26. doi: 10.1002/lary.70393. Epub ahead of print. PMID: 41588706.

[J.35] Dorr DA, Krussel A, Hauck R, Jackson C, Dalal A, Bedrick S, Payne PRO; **Bridge2AI-Voice Consortium**; Hersh W. Adapting data science competencies by role and purpose: Voice AI. ***Front Digit Health***. 2025 Oct 21;7:1610253. doi: 10.3389/fdgth.2025.1610253. PMID: 41195285; PMCID: PMC12582912.

[J.34] Rizzo A, Mozgai S, **Sigaras A**, Rubin JE, Jotwani R. Expert Consensus Best Practices for the Safe, Ethical, and Effective Design and Implementation of Artificially Intelligent Conversational Agent (i.e. Chatbot/Virtual Human) Systems in Healthcare Applications. ***Journal of Medical Extended Reality*** 2025 Sep 8 (Vol. 2, No. 1, pp. 209-222) doi: 10.1177/29941520251369450

[J.33] Jenkins P, Harrison R, Bedrick S, Karstens L; **Bridge2AI-Voice Consortium**; Hersh W. Voice as a biomarker: exploratory analysis for benign and malignant vocal fold lesions. ***Front Digit Health.*** 2025 Aug 12;7:1609811. doi: 10.3389/fdgth.2025.1609811. PMID: 40873766; PMCID: PMC12378753.

[J.32] Blatter A, Gallois H, Evangelista E, Bensoussan Y; **Bridge2AI-Voice Consortium**; Bélisle-Pipon JC. "Voice is the New Blood": a discourse analysis of voice AI health-tech start-up

websites. ***Front Digit Health.*** 2025 May 29;7:1568159. doi: 10.3389/fdgth.2025.1568159. PMID: 40510414; PMCID: PMC12158947.

[J.31] Anibal J, Doctor R, Boyer M, Newberry K, De Santiago I, Awan S, Abdel-Aty Y, Dion G, Daoud V, Huth H, Watts S, Wood BJ, Clifton D, Gelbard A, Powell M, Toghranegar J; **Bridge2AI Voice Consortium**; Bensoussan Y. Transformers for rapid detection of airway stenosis and stridor. ***Sci Rep***. 2025 May 2;15(1):15394. doi: 10.1038/s41598-025-99369-y. PMID: 40316652; PMCID: PMC12048665.

[J.30] Moothedan E, Boyer M, Watts S, Abdel-Aty Y, Ghosh S, Rameau A, **Sigaras A**, Elemento O; Bridge2AI-Voice Consortium; Bensoussan Y. The Bridge2AI-voice application: initial feasibility study of voice data acquisition through mobile health. ***Front Digit Health.*** 2025 Apr 15;7:1514971. doi: 10.3389/fdgth.2025.1514971. PMID: 40302934; PMCID: PMC12037532.

[J.29] Bhavsar N, Sriram S, Balasubramanian S, Davidson C, Piechowski W, Zimm J, Sanapala S, Fortenko A, Lame M, Greenwald PW, St. George J, **Sigaras A.** AIRWAY-XR: Augmented Instruction to Refine Wayfinding and Yielding Skills in Emergency Medicine Residents for Intubation using Mixed Reality Technology. ***medRxiv***. 2025 Jan 6:2025-01.

[J.28] Waltman P, Chandra P, Eng KW, Wilkes DC, Park H, Pabon C, Gorski K, Greco N, SimiM, Tang JM, Zisimopoulos P, King A, Al Assaad M, Teneyck T, Roberts D, Monge J, Demichelis F, Tam W, Ouseph MM, **Sigaras A**, Beltran H, Rennert H, Lindeman N, Song W, Solomon J, Mosquera JM, Kim R, Catalano J, Hassane DC, Sigouros M, Elemento O, Alonso A, Sboner A. EXaCT-2: An augmented and customizable oncology-focused whole exome sequencing platform. ***medRxiv***. 2025:2024.12.05.24318515. doi: 10.1101/2024.12.05.24318515.

[J.27] Anibal J, Gunkel J, Awan S, Huth H, Nguyen H, Le T, Bélisle-Pipon JC, Boyer M, Hazen L; **Bridge2AI Voice Consortium**; Bensoussan Y, Clifton D, Wood B. The doctor will polygraph you now. ***Npj Health Syst.*** 2024;1(1):1. doi: 10.1038/s44401-024-00001-4. Epub 2024 Dec 5. PMID: 39759269; PMCID: PMC11698301.

[J.26] Annabestani M, Sriram S, Caprio A, Janghorbani S, Wong SC, **Sigaras A**, Mosadegh B. High-fidelity pose estimation for real-time extended reality (XR) visualization for cardiac catheterization. ***Sci Rep.*** 2024 Nov 6;14(1):26962. doi: 10.1038/s41598-024-76384-z. PMID: 39505924; PMCID: PMC11542031.

[J.25] Annabestani M, Sriram S, Wong SC, **Sigaras A**, Mosadegh B. Advanced XR-Based 6-DOF Catheter Tracking System for Immersive Cardiac Intervention Training. ***arXiv preprint*** arXiv:2411.02611. 2024 Nov 4.

[J.24] Bahr R, Anibal J, Bedrick S, Bélisle-Pipon JC, Bensoussan Y, Blaylock N, Castermans J, Comito K, Dorr D, Hale G, Jackson C, Krussel A, Kuman K, Komarlu AR, Lerner-Ellis J, Powell M, Ravitsky V, Rameau A, Reavis C, **Sigaras A**, Cruz SS, Vojtech J, Urbano M, Watts S, Zhao R, Toghranegar J; Bridge2AI-Voice Consortium. Workshop summaries from the 2024 voice AI symposium, presented by the Bridge2AI-voice consortium. ***Front Digit Health***. 2024 Oct 30;6:1484818. doi: 10.3389/fdgth.2024.1484818. PMID: 39540145; PMCID: PMC11557516.

[J.23] Awan SN, Bahr R, Watts S, Boyer M, Budinsky R; **Bridge2AI-Voice Consortium**; Bensoussan Y. Evidence-Based Recommendations for Tablet Recordings From the Bridge2AI-Voice

Acoustic Experiments. ***J Voice***. 2024 Sep 20:S0892-1997(24)00283-2. doi: 10.1016/j.jvoice.2024.08.029. Epub ahead of print. PMID: 39306498; PMCID: PMC11922786.

[J.22] Rajendran S, Brendel M, Barnes J, Zhan Q, Malmsten JE, Zisimopoulos P, **Sigaras A**, Ofori-Atta K, Meseguer M, Miller KA, Hoffman D, Rosenwaks Z, Elemento O, Zaninovic N, Hajirasouliha I. Automatic ploidy prediction and quality assessment of human blastocysts using time-lapse imaging. ***Nat Commun***. 2024 Sep 5;15(1):7756. doi: 10.1038/s41467-024-51823-7. PMID: 39237547; PMCID: PMC11377764.

[J.21] Evangelista EG, Bélisle-Pipon JC, Naunheim MR, Powell M, Gallois H; **Bridge2AI-Voice Consortium**; Bensoussan Y. Voice as a Biomarker in Health-Tech: Mapping the Evolving Landscape of Voice Biomarkers in the Start-Up World. ***Otolaryngol Head Neck Surg***. 2024 Aug;171(2):340-352. doi: 10.1002/ohn.830. Epub 2024 Jun 1. PMID: 38822764.

[J.20] Bélisle-Pipon JC, Powell M, English R, Malo MF, Ravitsky V; **Bridge2AI–Voice Consortium**; Bensoussan Y. Stakeholder perspectives on ethical and trustworthy voice AI in health care. ***Digit Health***. 2024 Jul 23;10:20552076241260407. doi: 10.1177/20552076241260407. PMID: 39055787; PMCID: PMC11271113.

[J.19] Evangelista E, Kale R, McCutcheon D, Rameau A, Gelbard A, Powell M, Johns M, Law A, Song P, Naunheim M, Watts S, Bryson PC, Crowson MG, Pinto J; **Bridge2AI-Voice**; Bensoussan Y. Current Practices in Voice Data Collection and Limitations to Voice AI Research: A National Survey. ***Laryngoscope***. 2024 Mar;134(3):1333-1339. doi: 10.1002/lary.31052. Epub 2023 Dec 13. PMID: 38087983.

[J.18] Bélisle-Pipon JC, Ravitsky V; **Bridge2AI-Voice Consortium**; Bensoussan Y. Individuals and (Synthetic) Data Points: Using Value-Sensitive Design to Foster Ethical Deliberations on Epistemic Transitions. ***Am J Bioeth***. 2023 Sep;23(9):69-72. doi: 10.1080/15265161.2023.2237436. PMID: 37647464; PMCID: PMC11278678.

[J.17] Barnes J, Brendel M, Gao VR, Rajendran S, Kim J, Li Q, Malmsten JE, Sierra JT, Zisimopoulos P, **Sigaras A**, Khosravi P, Meseguer M, Zhan Q, Rosenwaks Z, Elemento O, Zaninovic N, Hajirasouliha I. A non-invasive artificial intelligence approach for the prediction of human blastocyst ploidy: a retrospective model development and validation study. ***Lancet Digit Health.*** 2023 Jan;5(1):e28-e40. doi: 10.1016/S2589-7500(22)00213-8. PMID: 36543475.

[J.16] van der Mijn JC, Eng KW, Chandra P, Fernandez E, Ramazanoglu S, **Sigaras A**, Oromendia C, Gudas LJ, Tagawa ST, Nanus DM, Faltas BF, Beltran H, Sternberg CN, Elemento O, Sboner A, Mosquera JM, Molina AM. The genomic landscape of metastatic clear cell renal cell carcinoma after systemic therapy. ***Mol Oncol.*** 2022 Mar 1. doi: 10.1002/1878-0261.13204. Epub ahead of print. PMID: 35231161.

[J.15] Brendel M, Getseva V, Assaad MA, Sigouros M, **Sigaras A**, Kane T, Khosravi P, Mosquera JM, Elemento O, Hajirasouliha I. Weakly-supervised tumor purity prediction from frozen H&E stained slides. ***EBioMedicine***. 2022 Jun;80:104067. doi: 10.1016/j.ebiom.2022.104067. Epub 2022 May 26. PMID: 35644123; PMCID: PMC9157012.

[J.14] Kadri S, Sboner A, **Sigaras A**, Roy S. Containers in Bioinformatics: Applications, Practical Considerations, and Best Practices in Molecular Pathology. ***J Mol Diagn***. 2022 May;24(5):442-454. doi: 10.1016/j.jmoldx.2022.01.006. Epub 2022 Feb 18. PMID: 35189355.

[J.13] Matrai CE, Ohara K, Eng KW, Glynn SM, Chandra P, Chatterjee-Paer S, Motanagh S, Mirabelli S, Kurtis B, He B, **Sigaras A**, Gupta D, Chapman-Davis E, Holcomb K, Sboner A, Elemento O, Ellenson LH, Mosquera JM. Molecular Evaluation of Low-grade Low-stage Endometrial Cancer With and Without Recurrence. ***Int J Gynecol Pathol***. 2022 May 1;41(3):207-219. doi: 10.1097/PGP.0000000000000798. Epub 2021 Sep 6. PMID: 34483300; PMCID: PMC9018213.

[J.12] Khosravi P, Lysandrou M, Eljalby M, Li Q, Kazemi E, Zisimopoulos P, **Sigaras A**, Brendel M, Barnes J, Ricketts C, Meleshko D, Yat A, McClure TD, Robinson BD, Sboner A, Elemento O, Chughtai B, Hajirasouliha I. A Deep Learning Approach to Diagnostic Classification of Prostate Cancer Using Pathology-Radiology Fusion. ***J Magn Reson Imaging***. 2021 Aug;54(2):462-471. doi: 10.1002/jmri.27599. Epub 2021 Mar 14. PMID: 33719168; PMCID: PMC8360022.

[J.11] Guo C, Crespo M, Gurel B, Dolling D, Rekowski J, Sharp A, Petremolo A, Sumanasuriya S, Rodrigues DN, Ferreira A, Pereira R, Figueiredo I, Mehra N, Lambros MBK, Neeb A, Gil V, Seed G, Terstappen L, Alimonti A, Drake CG, Yuan W, de Bono JS, Robinson D, Van Allen EM, Wu YM, Schultz N, Lonigro RJ, Mosquera JM, Montgomery B, Taplin ME, Pritchard CC, Attard G, Beltran H, Abida W, Bradley RK, Vinson J, Cao X, Vats P, Kunju LP, Hussain M, Tomlins SA, Cooney KA, Smith DC, Brennan C, Siddiqui J, Mehra R, Chen Y, Rathkopf DE, Morris MJ, Solomon SB, Durack JC, Reuter VE, Gopalan A, Gao J, Loda M, Lis RT, Bowden M, Balk SP, Gaviola G, Sougnez C, Gupta M, Evan YY, Mostaghel EA, Cheng HH, Mulcahy H, True LD, Plymate SR, Dvinge H, Ferraldeschi R, Flohr P, Miranda S, Zafeiriou Z, Tunariu N, Mateo J, Perez-Lopez R, Demichelis F, Robinson BD, Schiffman M, Nanus DM, Tagawa ST, **Sigaras A**, Eng KW, Elemento O, Sboner A, Heath EI, Scher HI, Pienta KJ, Kantoff P, Rubin MA, Nelson PS, Garraway LA, Sawyers CL, Chinnaiyan AM. CD38 in Advanced Prostate Cancers. ***European Urology***. 2021 Mar 5.
doi: 10.1016/j.eururo.2021.01.017

[J.10] Curchoe CL, Malmsten J, Bormann C, Shafiee H, Farias AFS, Mendizabal G, Chavez-Badiola A, **Sigaras A**, Alshubbar H, Chambost J, Jacques C, Pena CA, Drakeley A, Freour T, Hajirasouliha I, Hickman CFL, Elemento O, Zaninovic N, Rosenwaks Z. Predictive modeling in reproductive medicine: Where will the future of artificial intelligence research take us?. ***Fertility and Sterility*** 2020 Nov 1;114(5):934-40.
doi: 10.1016/j.fertnstert.2020.10.040

[J.9] Segal E, Zhang F, Lin X, King G, Shalem O, Shilo S, Allen WE, Alquaddoomi F, Altae-Tran H, Anders S, Balicer R, Bauman T, Bonilla X, Booman G, Chan AT, Cohen O, Coletti S, Davidson N, Dor Y, Drew DA, Elemento O, Evans G, Ewels P, Gale J, Gavrieli A, Geiger B, Grad YH, Greene CS, Hajirasouliha I, Jerala R, Kahles A, Kallioniemi O, Keshet A, Kocarev L, Landua G, Meir T, Muller A, Nguyen LH, Oresic M, Ovchinnikova S, Peterson H, Prodanova J, Rajagopal J, Rätsch G, Rossman H, Rung J, Sboner A, **Sigaras A**, Spector T, Steinherz R, Stevens I, Vilo J, Wilmes P. Building an international consortium for tracking coronavirus health status. ***Nat Med*** 26, 1161–1165 (2020).
doi: 10.1038/s41591-020-0929-x

[J.8] Khosravi P, Lysandrou M, Eljalby M, Brendel M, Li Q, Kazemi E, Barnes J, Zisimopoulos P, **Sigaras A**, Ricketts C, Meleshko D, Yat A, McClure T.D, Robinson B.D, Sboner A, Elemento O, Chughtai B, Hajirasouliha I. Biopsy-free prediction of prostate cancer aggressiveness using deep learning and radiology imaging. ***medRxiv***. 2019 Jan 1.
doi:10.1101/2019.12.16.19015057

[J.7] Khosravi P, Kazemi E, Zhan Q, Malmsten JE, Toschi M, Zisimopoulos P, **Sigaras A**, Lavery S, Cooper LA, Hickman C, Meseguer M. Deep learning enables robust assessment and selection of human blastocysts after in vitro fertilization. ***NPJ digital medicine***. 2019 Apr 4;2(1):1-9.
doi:10.1038/s41746-019-0096-y

[J.6] Sailer V, Eng KW, Zhang T, Bareja R, Pisapia D, **Sigaras A**, Bhinder B, Romanel A, Wilkes D, Sticca E, Cyrta C, Rao R, Sahota S, Pauli C, Beg S, Motanagh S, Kossai M, Fontunge J, Puca L, Rennert H, Xiang JZ, Greco N, Kim R, MacDonald TY, McNary T, Blattner-Johnson M, Schiffman MH, Faltas BM, Greenfield JP, Rickman D, Andreopoulou E, Holcomb K, Vahdat LT, Scherr DS, Koen van Besien, Barbieri CE, Robinson BD, Fine HA, Ocean AJ, Molina A, Shah MA, Nanus DM, Pan Q, Demichelis F, Tagawa ST, Song W, Mosquera JM, Sboner A, Rubin MA, Elemento O, and Beltran H Integrative Molecular Analysis of Patients With Advanced and Metastatic Cancer. ***JCO Precision Oncology*** 2019 :3, 1-12.
doi:10.1200/PO.19.00047

[J.5] Sailer V, Schiffman MH, Kossai M, Cyrta J, Beg S, Sullivan B, Pua BB, Lee KS, Talenfeld AD, Nanus DM, Tagawa ST, Robinson BD, Rao RA, Pauli C, Bareja R, Beltran LS, **Sigaras A**, Eng KW, Elemento O, Sboner A, Rubin MA, Beltran H, Mosquera JM. Bone biopsy protocol for advanced prostate cancer in the era of precision medicine. ***Cancer***. 2018 Mar 1;124(5):1008-1015.
doi: 10.1002/cncr.31173. Epub 2017 Dec 19.

[J.4] Pisapia DJ, Salvatore S, Pauli C, Hissong E, Eng K, Prandi D, Sailer VW, Robinson BD, Park K, Cyrta J, Tagawa ST, Kossai M, Fontugne J, Kim R, **Sigaras A**, Rao R, Pancirer D, Faltas B, Bareja R, Molina AM, Nanus DM, Rajappa P, Souweidane MM, Greenfield J, Emde AK, Robine N, Elemento O, Sboner A, Demichelis F, Beltran H, Rubin MA, Mosquera JM. Next-Generation Rapid Autopsies Enable Tumor Evolution Tracking and Generation of Preclinical Models. ***JCO Precis Oncol***. 2017;2017.
doi: 10.1200/PO.16.00038. Epub 2017 Jun 14.

[J.3] Bose R, Karthaus WR, Armenia J, Abida W, Iaquinta PJ, Zhang Z, Wongvipat J, Wasmuth EV, Shah N, Sullivan PS, Doran MG, Wang P, Patruno A, Zhao Y, Robinson D, Van Allen EM, Wu YM, Schultz N, Lonigro RJ, Mosquera JM, Montgomery B, Taplin ME, Pritchard CC, Attard G, Beltran H, Abida W, Bradley RK, Vinson J, Cao X, Vats P, Kunju LP, Hussain M, Tomlins SA, Cooney KA, Smith DC, Brennan C, Siddiqui J, Mehra R, Chen Y, Rathkopf DE, Morris MJ, Solomon SB, Durack JC, Reuter VE, Gopalan A, Gao J, Loda M, Lis RT, Bowden M, Balk SP, Gaviola G, Sougnez C, Gupta M, Yu EY, Mostaghel EA, Cheng HH, Mulcahy H, True LD, Plymate SR, Dvinge H, Ferraldeschi R, Flohr P, Miranda S, Zafeiriou Z, Tunariu N, Mateo J, Perez-Lopez R, Demichelis F, Robinson BD, Schiffman M, Nanus DM, Tagawa ST, **Sigaras A**, Eng KW, Elemento O, Sboner A, Heath EI, Scher HI, Pienta KJ, Kantoff P, de Bono JS, Rubin MA, Nelson PS, Garraway LA, Sawyers CL, Chinnaiyan AM, Zheng D, Schultz N, Sawyers CL. ERF mutations reveal a balance of ETS factors controlling prostate oncogenesis. ***Nature***. 2017 Jun 29;546(7660):671-675. Epub 2017 Jun 14. PubMed PMID: 28614298; PubMed Central PMCID: PMC5576182.
doi: 10.1038/nature22820

[J.2] Beltran H, Eng K, Mosquera JM, **Sigaras A**, Romanel A, Rennert H, Kossai M, Pauli C, Faltas B, Fontugne J, Park K, Banfelder J, Prandi D, Madhukar N, Zhang T, Padilla J, Greco N, McNary TJ, Herrscher E, Wilkes D, MacDonald TY, Xue H, Vacic V, Emde AK, Oschwald D, Tan AY, Chen Z, Collins C, Gleave ME, Wang Y, Chakravarty D, Schiffman M, Kim R, Campagne F, Robinson BD, Nanus DM, Tagawa ST, Xiang JZ, Smogorzewska A, Demichelis F, Rickman DS, Sboner A, Elemento O, Rubin MA. Whole-Exome Sequencing of Metastatic Cancer and Biomarkers of Treatment Response. ***JAMA Oncol***. 2015 Jul;1(4):466-74. doi: 10.1001/jamaoncol.2015.1313.

[J.1] Robinson D, Van Allen EM, Wu YM, Schultz N, Lonigro RJ, Mosquera JM, Montgomery B, Taplin ME, Pritchard CC, Attard G, Beltran H, Abida W, Bradley RK, Vinson J, Cao X, Vats P, Kunju LP, Hussain M, Feng FY, Tomlins SA, Cooney KA, Smith DC, Brennan C, Siddiqui J, Mehra R, Chen Y, Rathkopf DE, Morris MJ, Solomon SB, Durack JC, Reuter VE, Gopalan A, Gao J, Loda M, Lis RT, Bowden M, Balk SP, Gaviola G, Sougnez C, Gupta M, Yu EY, Mostaghel EA, Cheng HH, Mulcahy H, True LD, Plymate SR, Dvinge H, Ferraldeschi R, Flohr P, Miranda S, Zafeiriou Z, Tunariu N, Mateo J, Perez-Lopez R, Demichelis F, Robinson BD, Schiffman M, Nanus DM, Tagawa ST, **Sigaras A**, Eng KW, Elemento O, Sboner A, Heath EI, Scher HI, Pienta KJ, Kantoff P, de Bono JS, Rubin MA, Nelson PS, Garraway LA, Sawyers CL, Chinnaiyan AM. Integrative clinical genomics of advanced prostate cancer. ***Cell***. 2015 May 21;161(5):1215-1228.
doi: 10.1016/j.cell.2015.05.001.

Conference Proceedings

[C.1] A Reiter, **A Sigaras**, D Fowler, PK Allen. Surgical Structured Light for 3D minimally invasive surgical imaging. ***Intelligent Robots and Systems (IROS 2014), 2014 IEEE/RSJ International Conference on***. 2014 November 6, 1282 – 1287.
doi: 10.1109/IROS.2014.6942722

Abstracts

[A.16] Bales M, Wood EA, **Sigaras A**, Sholle EC, Imperato-McGinley J. 378 Leveraging large language models to communicate translational science benefits at Weill Cornell Medicine Clinical and Translational Science Center. ***Journal of Clinical and Translational Science***. 2025 Apr 1;9(s1):117-.

[A.15] Bhavsar N, Sriram S, Balasubramanian S, Davidson C, Fortenko A, Lame M, Greenwald P, George J, **Sigaras A**, Piechowski W, Sanapala S. 362 AIRWAY-MR: Assessing Traditional Intubation Didactics With a Novel Mixed Reality Module. ***Annals of Emergency Medicine.*** 2024 Oct 1;84(4):S165.

[A.14] Sboner A, Simi M, Gorski K, Delpe P, Fernandez E, Solomon J, Tang J, Zisimopoulos P, **Sigaras A.** An Event-Driven Architecture to Simplify Systems Communication in the Molecular Laboratory. ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** 2023 Nov 1 (Vol. 25, No. 11, pp. S78-S79). STE 800, 230 PARK AVE, NEW YORK, NY 10169 USA: ELSEVIER SCIENCE INC.

[A.13] Bhavsar N, Sriram S, Coleman D, Sulcov A, Fortenko A, Lame M, Greenwald P, George J, **Sigaras A**. 253 AIRWAY-XR: Augmented Instruction to Refine Wayfinding and Yielding Skills in Emergency Medicine Residents for Intubation Using Mixed Reality Technology. ***Annals of Emergency Medicine.*** 2023 Oct 1;82(4):S113-4.

[A.12] Perez N, Elemento O, Sboner A, **Sigaras A**. Building the Healthcare Metaverse: Leveraging Mixed Reality Guides to Augment Standard Operating Procedures (SOP) in a Precision Medicine Laboratory. ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** 2022 Nov 1 (Vol. 24, No. 10, pp. S85-S86). doi: 10.1016/S1525-1578(22)00284-7

[A.11] Sboner A, **Sigaras A**, Davis J, Roshal S, Wilkes D, Hebding C, Bockelman D, Kane T, Ackermann S, Sigouros M, Catalano J, Mosquera JM, Sternberg C, Loda M, Weinstein H, Song W, Elemento O. mrLab: Leveraging Mixed Reality in a Precision Medicine Laboratory to Increase Safety and Productivity of Healthcare Workers during the COVID-19 Pandemic. ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** 2020 Nov 1 (Vol. 22, No. 11, pp. S55-S56). doi: 10.1016/S1525-1578(20)30513-4

[A.10] Sboner A, Zisimopoulos P, Tang J, Oakley J, Spiewack M, Velu P, Solomon J, Song W, Park H, Kluk M, Mosquera J, Fernandez E, Simi M, Dorsaint P, Eng K, Catalano J, Pisapia D, Tran H, Tam W, Rennert H, Loda M, Elemento O, **Sigaras A**. Many NGS-Based Assays, One Platform: Ensuring a High-Quality Case Review and Sign-out Process with NGS Reporter (NGSR). ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** 2020 Nov 1 (Vol. 22, No. 11, pp. S54-S54) doi: 10.1016/S1525-1578(20)30513-4

[A.9] Sboner A, Sternberg C, Mosquera JM, Song W, Kluk M, Tam W, Rennert H, Pisapia D, Catalano J, Cheang G, Wilkes D, Bulaon D, Martin LM, **Sigaras A**, Eng K, Bareja R, Kim R, Loda M, Elemento O. Abstract IA33: Precision medicine at Weill Cornell Medicine/New York Presbyterian: Breaking silos, integrating resources, being inclusive **Cancer Epidemiol Biomarkers** Prev June 1 2020 (29) (6 Supplement 2) IA33; doi: 10.1158/1538-7755.DISP19-IA33

[A.8] Sboner A, Eng K, Zisimopoulus P, **Sigaras A**, Simi M, Ramazanoglu S, Spiewack M, Oakley J, Parmar S, Tang J, Catalano J, Kim R, Tam W, Kluk M, Song W, Elemento O. Building Scalable and Robust Informatics Systems for Clinical Genomic Assays in the Precision Medicine Era ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** (Vol. 22, No. 5, pp. S6-S6). STE 800, 230 PARK AVE, NEW YORK, NY 10169 USA: ELSEVIER SCIENCE INC.

[A.7] McIntire P, Ginter P, Eng KW, Lapolla D, **Sigaras A**, Beg S, Manohar J, Greco N, Zhang T, Bareja R, Sboner A, Elemento O, Andreopoulou E, Mosquera JM. Whole exome sequencing analysis of local/regional and distant metastatic breast carcinoma. ***USCAP 2020 Abstracts:*** Index of Abstract Authors. Mod Pathol 33, 2097–2136 (2020) doi:10.1038/s41379-020-0486-3

[A.6] **Sigaras A**, Wilkes D, Hebding C, Martin LM, Bulaon D, Catalano J, Mosquera JM, Sternberg C, Loda M, Weinstein H, Song W, Elemento O, Sboner A. Mixed Reality for a Precision Medicine Laboratory: The Future Is Now!. ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** 2019 Nov 1 (Vol. 21, No. 6, pp. 1172-1172). doi:10.1016/S1525-1578(19)30391-5

[A.5] Uppal M, **Sigaras A**, Zisimopoulos P, Tang J, Sternberg C.N, Song W, David Pisapia D, Rennert H, Sboner A. A crowdsourcing approach for high-quality interpretation of cancer variants ***ICGC 15th Scientific Workshop and 2nd ARGO Meeting*** May 28 2019

[A.4] **Sigaras A**, Eng KW, Sticca E, Ramazanoglu S, Parmar S, Aljaludi A, Israel M, Artz D, Leslin C, D'Agoustine J, Cheang G. Integrating Clinical Genomics into Electronic Health Records to Foster Precision Medicine. ***In JOURNAL OF MOLECULAR DIAGNOSTICS*** 2018 Nov 1 (Vol. 20, No. 6, pp. 965-965). doi:10.1016/S1525-1578(18)30401-X

[A.3] J Gao, T Mazor, E Ciftci, P Raman, P Lukasse, I Bahceci, **A Sigaras**, A Abeshouse, I de Bruijn, B Gross, R Kundra, A Lisman, A Ochoa, R Sheridan, J Su, S O. Sumer, Y Sun, A Wang, J Wang, M Wilson, H Zhang, P Kumari, J Lindsay, K Kalletla, K Zhu, O Plantalech, F Schaeffer, S Tan, D Zaal, S van Hagen, K van Bochove, U Dogrusoz, T Pugh, A Resnick, C Sander, N Schultz, E Cerami Abstract 923: The cBioPortal for Cancer Genomics: An intuitive opensource platform for exploration, analysis and visualization of cancer genomics data. ***Proceedings: AACR Annual Meeting 2018***, Volume 78, Issue 13, April 14-18, 2018; Chicago, IL, doi: 10.1158/1538-7445.AM2018-923

[A2] J Catalano, G Cheang, D Pancirer, E Merzier, Y Li, H Tran, **A Sigaras**. 331 The Development of a Custom LIMS: An Introduction and Guide to Successful Implmentation. ***American Journal of Clinical Pathology***, Volume 149, Issue suppl_1, 11 January 2018, Pages S142–S143, doi: 10.1093/ajcp/aqx127.330

[A1] H Beltran, K Eng, J Mosquera, **A Sigaras**, A Romanel, H Rennert, M Kossai, C Pauli, B Faltas, J Fontugne, B Robinson, D Nanus, S Tagawa, J Xiang, F Demichelis, D Rickman, A Sboner, O Elemento and M Rubin. Precision cancer medicine program for whole-exome sequencing of metastatic tumors reveals biomarkers of response. ***Cancer Research***, Volume 75, Issue 15, 1 August 2015, Page 4745 doi: 0.1158/1538-7445.AM2015-4745

Posters

[P.10] Perez N, Elemento O, Sboner A, **Sigaras A**. Building the Healthcare Metaverse: Leveraging Mixed Reality Guides to Augment Standard Operating Procedures (SOP) in a Precision Medicine Laboratory. ***AMP 2022 Annual Meeting & Expo,*** 5 November 2020

[P.9] Sboner A, **Sigaras A**, Davis J, Roshal S, Wilkes D, Hebding C, Bockelman D, Kane T, Ackermann S, Sigouros M, Catalano J, Mosquera JM, Sternberg C, Loda M, Weinstein H, Song W, Elemento O. mrLab: Leveraging Mixed Reality in a Precision Medicine Laboratory to Increase Safety and Productivity of Healthcare Workers during the COVID-19 Pandemic. ***AMP 2020 Annual Meeting & Expo,*** 19 November 2020

[P.8] Sboner A, Zisimopoulos P, Tang J, Oakley J, Spiewack M, Velu P, Solomon J, Song W, Park H, Kluk M, Mosquera J, Fernandez E, Simi M, Dorsaint P, Eng K, Catalano J, Pisapia D, Tran H, Tam W, Rennert H, Loda M, Elemento O, **Sigaras A**. Many NGS-Based Assays, One Platform: Ensuring a High-Quality Case Review and Sign-out Process with NGS Reporter (NGSR). ***AMP 2020 Annual Meeting & Expo,*** 19 November 2020

[P.7] McIntire P, Ginter P, Eng KW, Lapolla D, **Sigaras A**, Beg S, Manohar J, Greco N, Zhang T, Bareja R, Sboner A, Elemento O, Andreopoulou E, Mosquera JM. Whole exome sequencing analysis of local/regional and distant metastatic breast carcinoma. ***USCAP 2020 Annual Meeting,*** 3 March 2020

[P.6] **Sigaras A**, Wilkes D, Hebding C, Martin LM, Bulaon D, Catalano J, Mosquera JM, Sternberg C, Loda M, Weinstein H, Song W, Elemento O, Sboner A. Mixed Reality for a Precision Medicine Laboratory: The Future Is Now!. ***AMP 2019 Annual Meeting & Expo***, 9 November 2019

[P.5] Uppal M, **Sigaras A**, Zisimopoulos P, Tang J, Sternberg C.N, Song W, David Pisapia D, Rennert H, Sboner A. A crowdsourcing approach for high-quality interpretation of cancer variants ***ICGC 15th Scientific Workshop and 2nd ARGO Meeting*** May 28 2019

[P.4] **A Sigaras**, K Eng, E Sticca, S Ramazanoglu, S Parmar, A Aljaludi, M Israel, D Artz, C Leslin, J D'Augustine, G Cheang, D Pancirer, J Tang, P Zisimopoulos, M Rubin, W Tam, M Kluk, H Rennert, J Catalano, R Kim, H Beltran, O Elemento, W Song, A Sboner. Integrating Clinical Genomics into Electronic Health Records to Foster Precision Medicine. ***AMP 2018 Annual Meeting & Expo***, 2 November 2018

[P.3] **A Sigaras**, S Roshal, A Sboner, M Rubin, O Elemento. Healthcare Applications for Immersive Remote Collaboration of 3D Medical Data using Virtual and Mixed Reality. ***2017 Startup Symposium***, Weill Cornell Medicine, 26 January 2017

[P.2] E Vinolo, D Alférez, F Amant, D Annibali, J Arribas, M Bentires-Alj, C Bernadó, A Bertotti, A Biankin, A Bruna, E Budinská, A Byrne, C Caldas, O Casanovas, D K. Chang, R B. Clarke, S Corso, G Coukos, V Dangles-Marie, D Decaudin, J Depreeuw, Z Dudová, O Elemento, S Giordano, E Gonzalez-Suarez, H Hafsi, E Hermans, M Hidalgo, G Inghirami, M Jarzabek, S de Jong, J Jonkers, K Kemper, A Křenek, M Kuba, L Lanfrancone, P López Casas, G Mælandsmo, E Marangoni, E Medico, I Miller, K Moran-Jones, B Morancho, F Nematti, J Henrik Norum, H Palmer, D Peeper, P Pelicci, A Piris-Giménez, M Pujana, S Roman, O Rueda, J Seoane, V Serra, **A Sigaras,** L Soucek, S Tejpar, M Tomas, L Trusolino, A van der Zee, M van de Ven; D Vanhecke, A Villanueva, B Wisman. The EurOPDX Consortium: Objectives, Achievements & Future Directions ***1st EurOPDX Workshop***, Switzerland, 3 October 2016

[P.1] A Reiter, **A Sigaras** and P Allen. Surgical Structured Light (SSL) for Real-Time Minimally-Invasive 3D Imaging. ***John Jones Symposium, Columbia University***, 10 May 2013

Talks, Lectures

Invited Talks

03/05/2026 **FutureCare: How AI + XR Are Reimagining Health and Healing**
First Thursdays
Next Jump NYC HQ

07/17/2025 **AI & Medical Extended Reality in the era of Precision Surgery**
Society of Robotic Surgery (SRS) 2025 Annual Meeting – Strasbourg, France
Palais du Congress Convention Center - Marie Curie B

06/09/2024 **Precision Healthcare - AI and XR interventions**
2024 PHIE - Transforming Health Professional Education, Research and Care, São Paulo, Brazil

06/30/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
36th Annual Congress of the Hellenic Neurosurgical Society & 16th Annual Neurosurgery Nurses Meeting, Athens Greece

10/01/2022 **AI and XR opportunities in Clinical Education and Care Delivery**
Virtual Healthcare in the Mainstream Symposium & Research Forum 2022
Weill Cornell Medicine

03/18/2021 **Metaverse: The Next Frontier in Medicine?**
Institute for Computational Biomedicine, Research in Progress Seminar Series
Weill Cornell Medicine

06/29/2021 **XR in Medicine in the era of Precision Medicine**
Stanford Healthcare Innovation Lab

05/16/2021 **XR in Medicine in the era of Precision Medicine**
9th International Summer School in Medical Biosciences Research & Management
World Hellenic Biomedical Association

01/15/2021 **XR in Medicine in the era of Precision Medicine**
Institute for Computational Biomedicine, Research in Progress Seminar Series
Weill Cornell Medicine

07/24/2020 **XR in Medicine**
2nd Biomedical Big Data Symposium
New York City College of Technology, City University of New York

02/14/2020 **AI and Mixed Reality in Precision Medicine**
2nd Biomedical Big Data Symposium
New York City College of Technology, City University of New York

01/31/2020 **Mixed Reality in Precision Medicine**
NYU Tandon School of Engineering, New York, NY

10/28/2019 – 10/29/2019 **Digital Technologies for Clinical Care** – Panelist
Artificial Intelligence for Cancer – Discussion Leader
Precision Medicine Research Symposium
Cornell Tech and Weill Cornell Medicine, New York, NY

10/25/2019 – 10/26/2019 **XR in Healthcare**
R Lab Master Class 2019

06/05/2019 **XR in Medicine in the era of Precision Medicine**
R Lab Well 2019

06/19/2019 **Big Data Visualization and Spatial Computing in Precision Medicine**
Biomedical Big Data Summer Bootcamp 2019
Weill Cornell Medicine

02/15/2019 **Big Data in Precision Medicine**
1st Biomedical Big Data Symposium
New York City College of Technology, City University of New York

06/20/2018 **Big Data Visualization - Mixed Reality, Virtual Reality and Augmented Reality**
Biomedical Big Data Summer Bootcamp 2018
Weill Cornell Medicine

01/11/2017 **How HoloLens transforms Healthcare**
Microsoft Reactor, Grand Central Tech, NY

11/21/2008 **Introduction to Robotics with MS Robotics Studio**
Microsoft, Athens, Greece

01/29/2008 **Robotics Warehouse Project**
Selected presentation/talk to the Chairman of Microsoft, Bill Gates and Kostas Karamanlis, Prime Minister of Greece at the inaugural opening of the Microsoft Innovation Center in Greece
Microsoft Innovation Center, Greece

12/17/2007 **Introduction to Robotics with MS Robotics Studio and LEGO Mindstorms NXT**
IEEE hosted event
University of Patras, Greece

Guest Lectures

02/18/2026 **AI and Medical Extended Reality to Care, Discover, and Teach**
CMBP 5005: Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

02/27/2025 **Designing for AI and Medical Extended Reality in the Era of Precision Medicine**
INFO 5358: 3D Interaction Design, Cornell Tech
Host: Prof. Harald Haraldsson

02/05/2025 **AI and Medical Extended Reality in the Era of Precision Medicine**
CMBP 5005: Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

03/06/2024 **AI and Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof. Eugenia Giannopoulou

01/31/2024 **AI and Medical Extended Reality in the Era of Precision Medicine**
Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

11/29/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof. Pegah Khosravi

03/01/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof: Pegah Khosravi

02/01/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

12/05/2022 **AI and Medical Extended Reality in the Era of Precision Medicine**
Topics in Pathophysiology, Deree College, The American College of Greece
Host: Prof: Alexia Victoria Polissidis

02/16/2022 **Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof. Eugenia Giannopoulou

11/17/2021 **Medical Extended Reality in the Era of Precision Medicine**
Data Structures and Algorithms for Computational Biology, Weill Cornell Medicine

Host: Prof. Iman Hajirasouliha

10/13/2021 **Medical Extended Reality in the Era of Precision Medicine**
Virtual and Augmented Reality, Cornell Tech
Host: Prof. Harald Haraldsson

12/04/2020 **XR in Medicine in the era of Precision Medicine**
Hostos Community College, City University of New York
Hosts: Prof. Yoel Rodriguez, Prof Anna Ivanova

11/21/2019 **Mixed Reality in Precision Medicine**
Virtual and Augmented Reality, Cornell Tech
Host: Prof. Harald Haraldsson

12/2/2015 **Precision Medicine**
Bioinformatics II, New York City College of Technology, City University of New York
Host: Prof. Eugenia Giannopoulou

11/24/2011 **Intro to Programming in C++ and C#**
Introduction to Programming, University of Piraeus, Greece
Host: Prof. Ioannis-Christos Panagiotopoulos

12/16/2010 **Introduction to Expression Blend**
Human Computer Interaction, University of Piraeus, Greece
Host: Prof. Maria Virvou

12/09/2010 **Introduction to Sketchflow**
E-learning, University of Piraeus, Greece
Host: Prof. Symeon Retalis

Selected Press and Media Coverage

03/19/2026 **Weill Cornell News**, Students Design AI-Powered Health Solutions at Cornell Health Hackathon.
https://news.weill.cornell.edu/news/2026/03/students-design-ai-powered-health-solutions-at-cornell-health-hackathon

09/16/2025 **Weill Cornell Medicine Enterprise Innovation News,** *Exploring Human Voice as a Biomarker*
https://innovation.weill.cornell.edu/news-events/exploring-human-voice-biomarker

07/01/2024 **Impact Weill Cornell Medicine Magazine, Summer 2024,** *Features – The Sounds of Science*
http://impact.weill.cornell.edu/summer-2024/features/sounds-science

12/19/2022 **Weill Cornell News**, *Harnessing Artificial Intelligence Technology for IVF Embryo Selection.*
https://news.weill.cornell.edu/news/2022/12/harnessing-artificial-intelligence-technology-for-ivf-embryo-selection

09/13/2022 **NIH News Releases -** *NIH launches Bridge2AI program to expand the use of artificial intelligence in biomedical and behavioral research.*
https://www.nih.gov/news-events/news-releases/nih-launches-bridge2ai-program-expand-use-artificial-intelligence-biomedical-behavioral-research

09/13/2022 **USF Health News**, *USF Health, Weill Cornell Medicine Earn Inaugural Funding in NIH's Newly Launched Bridge2AI Initiative, will Create Artificial Intelligence Platform for Using Voice to Diagnose Disease.*
https://hscweb3.hsc.usf.edu/blog/2022/09/13/usf-health-cornell-earns-inaugural-nih-funding-to-create-artificial-intelligence-platform-for-using-voice-to-diagnose-disease/

09/13/2022 **Weill Cornell News**, *USF Health, Weill Cornell Medicine Earn Inaugural Funding in NIH's Newly Launched Bridge2AI Initiative, will Create Artificial Intelligence Platform for Using Voice to Diagnose Disease.*
https://news.weill.cornell.edu/news/2022/09/usf-health-weill-cornell-medicine-earn-inaugural-funding-in-nih%E2%80%99s-newly-launched

03/07/2022 **Bloomberg***, The Next Life-Saving Innovation in Medicine May Be Developed Only for You.*
https://sponsored.bloomberg.com/article/weill-cornell-medicine/the-next-life-saving-innovation-in-medicine-may-be-developed-only-for-you

08/30/2021 **New York Times***, Experts still aren't sure about corporate meetings taking place in the metaverse.*
https://www.nytimes.com/live/2021/08/30/business/economy-stock-market-news#experts-still-arent-sure-about-corporate-meetings-taking-place-in-the-metaverse

01/30/2021 **NBC News, Peacock - The Overview,** *What Does the Future of Work Look Like?*

https://www.peacocktv.com/watch/playback/vod/GMO_00000000363812_01/b2c4bc2c-dafc-3391-900b-fde591ea42ef

05/28/2020 **Protocol,** *Coronavirus sent us home. Will VR bring us back together?* https://www.protocol.com/vr-headset-future-of-work

04/06/2020 **Macaulay Honors College News and Events***, Macaulay Honors College Students to Assist With COVID-19 Effort* https://macaulay.cuny.edu/news-and-events/macaulay-honors-college-students-to-assist-with-covid-19-effort/

04/01/2020 **New York Times***, How Are You Feeling? Surveys Aim to Detect Covid-19 Hot Spots Early.* https://www.nytimes.com/2020/04/01/world/middleeast/coronavirus-detection-surveys.html

07/15/2019 **CNN,** *Would you trust an algorithm to diagnose an illness?* https://www.cnn.com/2019/07/15/business/artificial-intelligence-healthcare/index.html

04/24/2019 **NYP - Health Matters,** *Behind the Latest Advance in IVF Treatment* https://healthmatters.nyp.org/behind-the-latest-advance-in-ivf-treatment/

04/11/2019 **Cornell Chronicle,** *Researchers test using AI to optimize IVF embryo selection* https://news.cornell.edu/stories/2019/04/researchers-test-using-ai-optimize-ivf-embryo-selection

04/05/2019 **Wired,** *AI Could Scan IVF Embryos to Help Make Babies More Quickly* https://www.wired.com/story/ai-could-scan-ivf-embryos-to-help-make-babies-more-quickly/

04/05/2019 **AI in Healthcare,** *AI optimizes embryo selection for IVF* https://www.aiin.healthcare/topics/imaging/ai-optimizes-embryo-selection-ivf

04/04/2019 **Wall Street Journal,** *IVF Often Doesn't Work. Could an Algorithm Help?* https://www.wsj.com/articles/ivf-often-doesnt-work-could-an-algorithm-help-11554386243

04/04/2019 **Weill Cornell News**, *Artificial Intelligence Approach Optimizes Embryo Selection for IVF* https://news.weill.cornell.edu/news/2019/04/artificial-intelligence-approach-optimizes-embryo-selection-for-ivf

04/04/2019 **Science Daily**, *Artificial intelligence approach optimizes embryo selection for IVF* https://www.sciencedaily.com/releases/2019/04/190404161003.htm

03/09/2018 **Microsoft Developer Blog**, *Voronoi Selection for Cancer Drug Network Visualization in Mixed Reality* https://www.microsoft.com/developerblog/2018/03/09/voronoi-selection-cancer-drug-network-visualization-mixed-reality/

10/17/2017	**Channel 9**, *Behind the Scenes: How Weill Cornell Medicine built a chatbot for clinicians to gain fast access to medical data* https://channel9.msdn.com/Blogs/DevRadio/DR1747
09/13/2017	**Microsoft Windows Blogs**, *Making mixed reality: a conversation with Alexandros Sigaras and Sophia Roshal* https://blogs.windows.com/windowsexperience/2017/09/13/making-mixed-reality-conversation-alexandros-sigaras-sophia-roshal/#B0hV1JuJsZoIfG2Q.97
06/15/2017	**Microsoft Technical Case Studies**, *Bot gives Weill Cornell clinicians fast access to medical data* https://microsoft.github.io/techcasestudies/bot%20framework/2017/06/15/WeillCornell.html
06/14/2017	**The Sociable**, *How Mixed Reality is Transforming Collaborative Cancer Research* https://sociable.co/technology/mixed-reality-cancer/
03/26/2017	**Arirang, Special Documentary** *Smart with Heart* https://www.youtube.com/watch?v=gHa4ddmdCUU
Winter 2016	**Weill Cornell Medicine Magazine, Volume 15, Number 3,** *Virtual Vision – 3D technology could offer a potent weapon in cancer care* http://www.weillcornellmedicine-digital.com/weillcornellmedicine/vol_15_no_3?pg=20#pg20
01/27/2016	**ABC7 Eyewitness News**, *New 3D goggles help doctors search for mutations in cancerous tumors* http://abc7ny.com/health/new-3d-goggles-help-doctors-search-for-mutations-in-cancerous-tumors/1176716/
07/15/2016	**Weill Cornell News**, *Researchers are Using Virtual Reality to Help Treat Cancer* https://weillcornell.org/news/researchers-are-using-virtual-reality-to-help-treat-cancer
07/13/2016	**CBS New York**, *Seen At 11: Doctors Using Virtual Reality To Treat Cancer Patients* http://newyork.cbslocal.com/2016/07/13/virtual-reality-cancer-patients/
02/24/2016	**Popular Science**, *Here's How Virtual Reality Could Help Doctors Treat Cancer* https://www.popsci.com/how-virtual-reality-could-help-doctors-treat-cancer
05/21/2015	**Science Daily**, *Scientists unveil prostate cancer's 'Rosetta Stone'* https://www.sciencedaily.com/releases/2015/05/150521133732.htm